

Configure API-KEY

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1. Course Content

- Using the previously registered API key, configure the API key for the car.

[!WARNING]

Note: Please ensure the car system is connected to the internet to use the cloud-based model services.

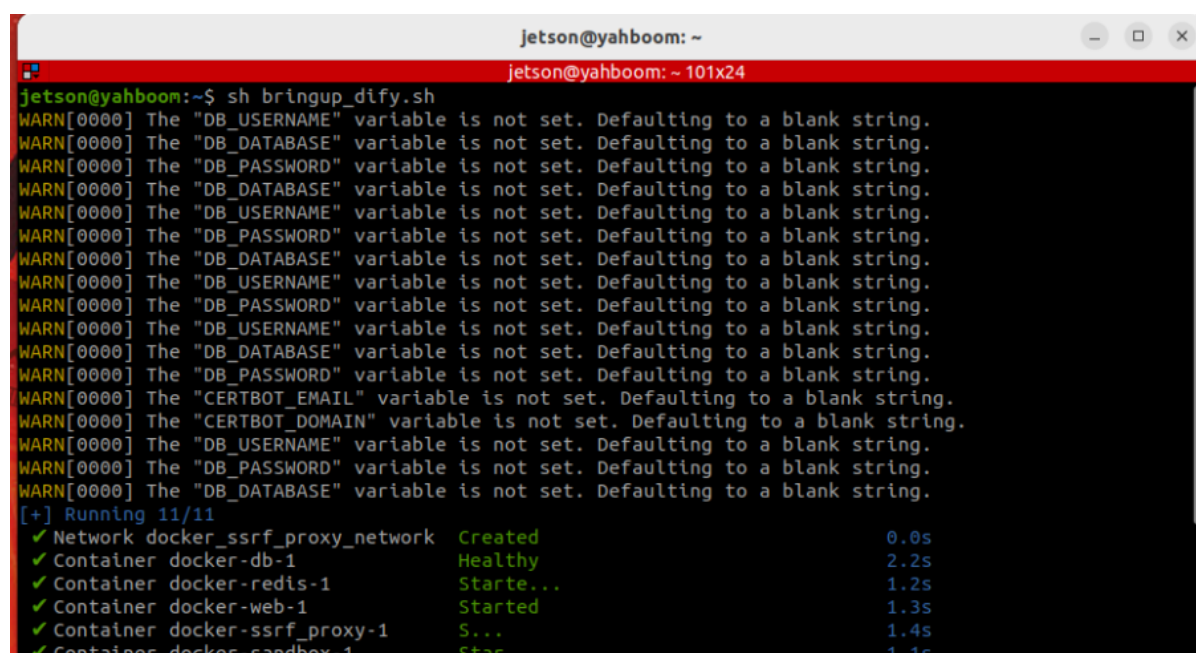
2. Start the Dify Service

[!TIP]

ROSMaster-A1 uses Dify to build a multi-agent system, with Dify centrally managing the access to cloud-based models.

- Connect to the car's infotainment system via VNC or SSH. Enter the following command in the terminal:

```
sh bringup_dify.sh
```



```
jetson@yahboom: ~  
jetson@yahboom: ~ 101x24  
jetson@yahboom:~$ sh bringup_dify.sh  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "CERTBOT_EMAIL" variable is not set. Defaulting to a blank string.  
WARN[0000] The "CERTBOT_DOMAIN" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
[+] Running 11/11  
✓ Network docker_ssrf_proxy_network Created 0.0s  
✓ Container docker-db-1 Healthy 2.2s  
✓ Container docker-redis-1 Started... 1.2s  
✓ Container docker-web-1 Started 1.3s  
✓ Container docker-ssrf_proxy-1 S... 1.4s  
✓ Container docker-sandbox-1 Star 1.1s
```

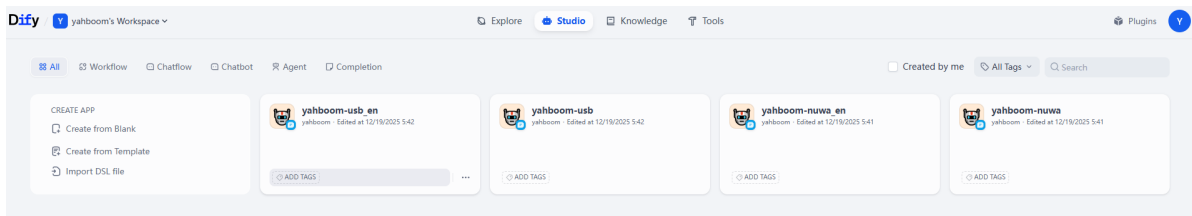
- View the infotainment system's IP address. This can be done via the OLED screen or in the terminal using `ifconfig`.

```
jetson@yahboom: ~  
jetson@yahboom: ~ 101x24  
ether 3a:ea:17:8a:52:89 txqueuelen 0 (Ethernet)  
RX packets 0 bytes 0 (0.0 B)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 30 bytes 2750 (2.7 KB)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
vethf5fe159: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
inet6 fe80::2c3f:efff:fed1:4939 prefixlen 64 scopeid 0x20<link>  
ether 2e:3f:ef:d1:49:39 txqueuelen 0 (Ethernet)  
RX packets 0 bytes 0 (0.0 B)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 22 bytes 3074 (3.0 KB)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
wlP1p1s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
inet 192.168.14.101 netmask 255.255.255.0 broadcast 192.168.14.255  
inet6 fe80::20e1:9dde:bc30:5479 prefixlen 64 scopeid 0x20<link>  
ether 48:8f:4c:d5:50:76 txqueuelen 1000 (Ethernet)  
RX packets 2765 bytes 2720194 (2.7 MB)  
RX errors 0 dropped 24 overruns 0 frame 0  
TX packets 1440 bytes 1410350 (1.4 MB)  
TX errors 0 dropped 2 overruns 0 carrier 0 collisions 0  
  
jetson@yahboom:~$
```

- Enter the infotainment system's IP address directly into your browser's address bar to access the Dify management page. If this is the first time logging in, you will need to use a username and password. You can select the language in the upper left corner.

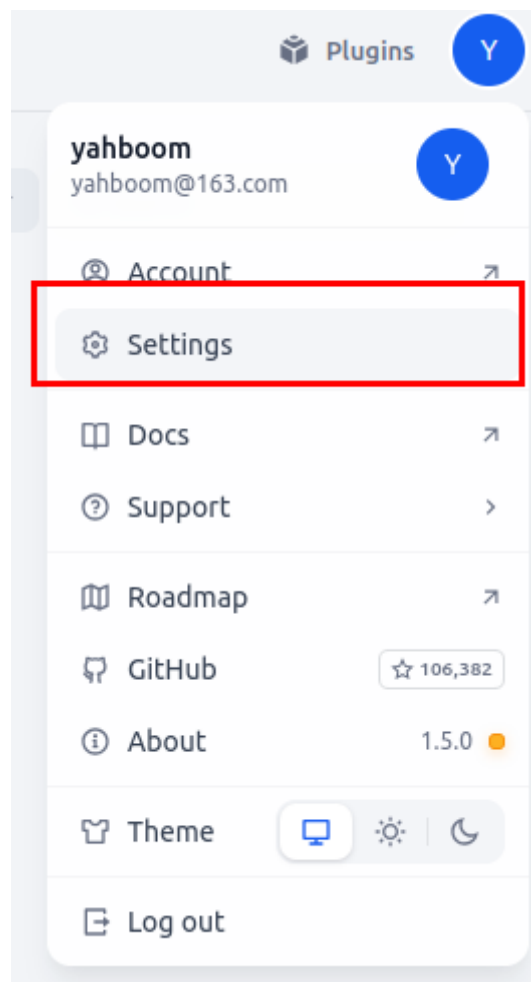
[!NOTE]

- Username: `yahboom@163.com`
- Password: `yahboom123`
- All account passwords, agent applications, and RAG data are stored locally.
- After logging in, see the image below.

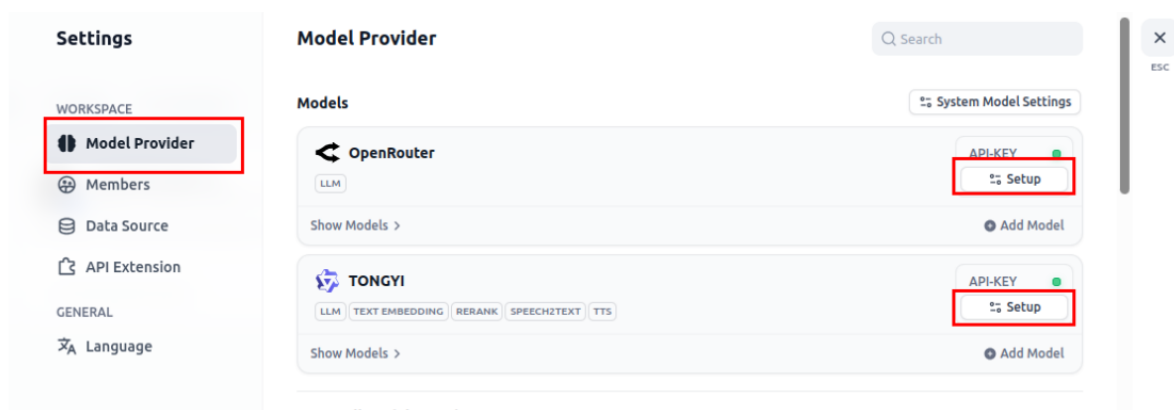


3. Configure Model Service Provider API-KEY

- Click Settings



- Here, we take configuring the API for a Bailian account as an example. Click Model Provider -> Setup



- Enter your Bailian platform's API-KEY, then select whether it is an international battle account, and click Save.

Setup TONGYI

API Key *

.....

Use International Endpoint *

☒ True

☐ False

[Get your API key from AliCloud](#)

Remove

Cancel

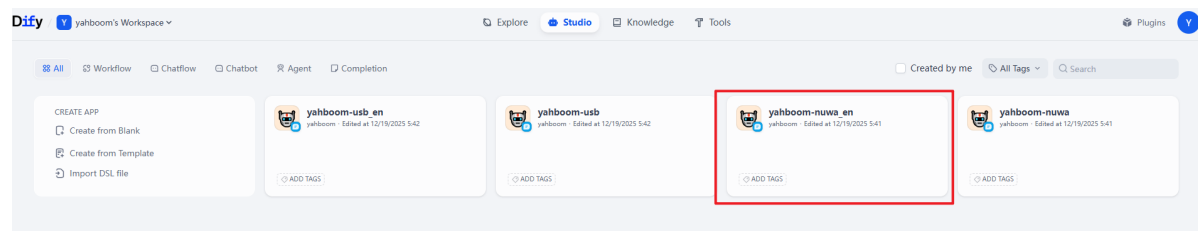
Save

Your API KEY will be encrypted and stored using PKCS1_OAEP technology.

4. Testing the API Key

[!TIP]

- If you need to test if the API key is available, you can refer to this tutorial. If not, you can ignore this section.
- Click on the yahboom-nuwa application in the studio (international clients can use the yahboom-nuwa_en application for testing).



- Click on "Decision-making AI" and "Executive-level AI," then select any model from the available options to test.



The screenshot displays the Qwen AI web interface. On the left, a 'MODEL' dropdown menu is open, showing a search bar and a list of models under the 'TONGYI' provider. The models listed include 'qvq-max-latest', 'qvq-max', 'qwq-plus-0305', 'qwq-plus', 'qwq-32b', 'DeepSeek-R1', 'DeepSeek-R1-Distill-Qwen-14B', 'DeepSeek-R1-Distill-Qwen-32B', 'DeepSeek-V3.2-exp', 'DeepSeek-V3.1', 'DeepSeek-V3', and 'qwen3-max'. The 'qwen3-max-2025-09-23' model is currently selected in the main interface. The main interface shows the title 'Decision-making AI' and a 'Preview' button. Below the title, there are tabs for 'SETTINGS' and 'LAST RUN'. The 'MODEL' section shows the selected model 'qwen3-max-2025-09-23' with a 'CHAT' button. The 'CONTEXT' section shows a snippet of code: 'Decision-making layer example lib... / (x) r... Array[Objj...'. The 'SYSTEM' section contains a prompt: 'You need to plan a reasonable sequence of task steps for the robot based on the user's input instructions. Workflow: 1. Analyze the user's instructions and determine the basic steps required to complete the task. 2. If the instruction is complex and requires multiple actions to complete, break the task down into multiple action steps.' The 'USER' section shows a prompt: '1167 Jinja (x) [trash] [copy] [paste]'.

MODEL

qwen3-max-2025-09-23 CHAT

Search model

TONGYI

- qvq-max-latest
- qvq-max
- qwq-plus-0305
- qwq-plus
- qwq-32b
- DeepSeek-R1
- DeepSeek-R1-Distill-Qwen-14B
- DeepSeek-R1-Distill-Qwen-32B
- DeepSeek-V3.2-exp
- DeepSeek-V3.1
- DeepSeek-V3
- qwen3-max

Model Provider Settings

Decision-making AI

Preview

Add description...

SETTINGS **LAST RUN**

MODEL

qwen3-max-2025-09-23 CHAT

CONTEXT

Decision-making layer example lib... / (x) r... Array[Objj...

SYSTEM

336 Jinja (x) [trash] [copy] [paste]

You need to plan a reasonable sequence of task steps for the robot based on the user's input instructions.

Workflow:

1. Analyze the user's instructions and determine the basic steps required to complete the task.
2. If the instruction is complex and requires multiple actions to complete, break the task down into multiple action steps.

USER

1167 Jinja (x) [trash] [copy] [paste]

- Click Preview, then type "Hello" and press Enter

Plugins Y

Preview Publish Features

1

PREVIEW

robot_feedback (optional)

False

DECISION-MAKING AI

ven3-max-2025-0... CHAT

ANSWER

① Executiv... [x] structured out...

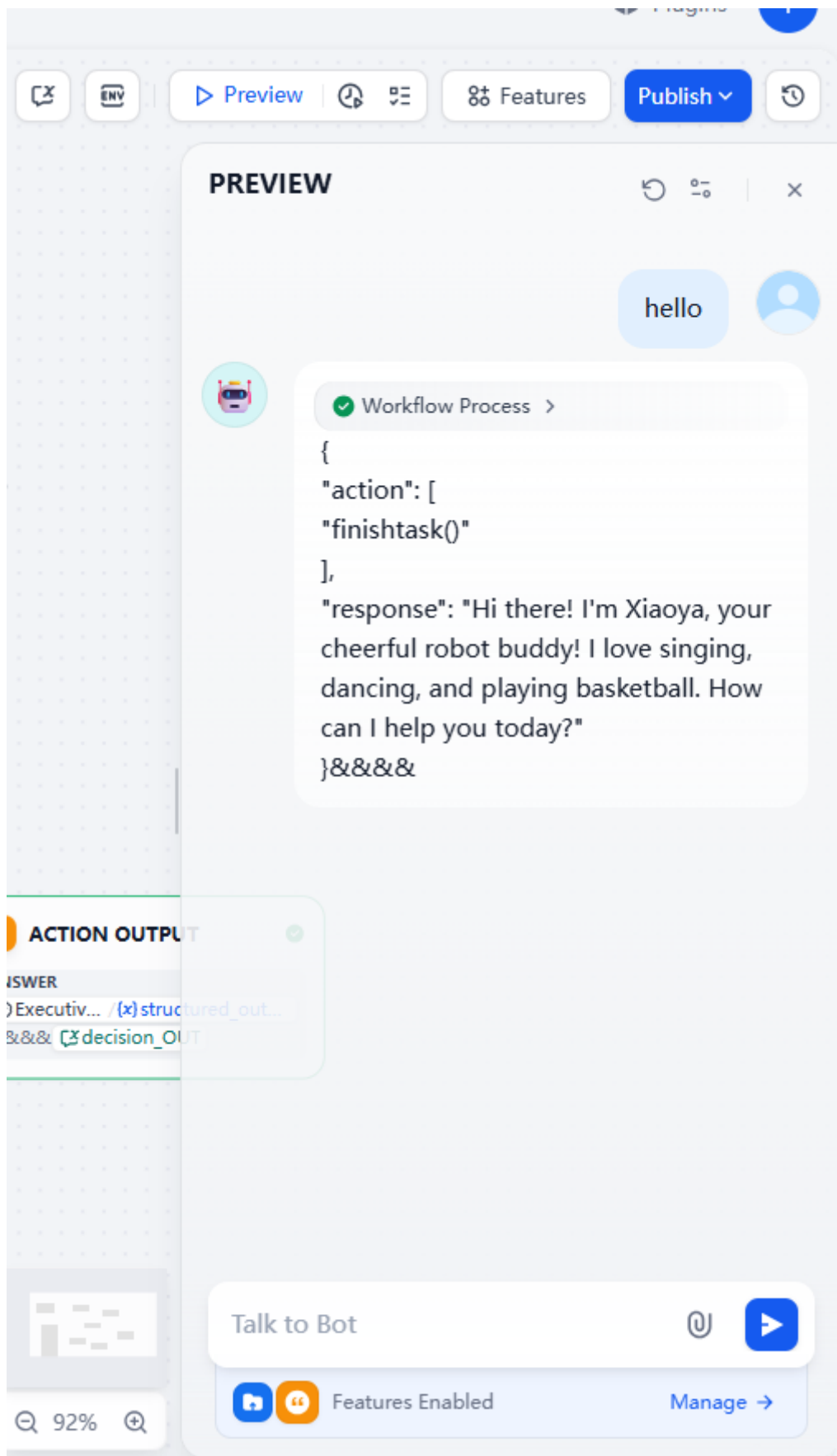
8&8&8 [x] decision OUT

2

Hello

Features Enabled Manage →

You can see the agent's reply



5. Configuring the multi_brains

- Enter the parameter file path

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/config/multi_brains_setting.
yaml
```

- Open the parameter file (using any method such as vscode, nano command line, or VNC), and make the following modifications in the multi_brains_setting.yaml file:
 1. Change the `LANGUAGE` parameter to en
 2. Select the corresponding application for the camera
 3. Change `ASR_SUPPLIER` in ASR function setting to xunfei
 4. Change `TTS_SUPPLIER` in TTS function setting to xunfei

```

0: > | multi_brains_setting.yaml
1 #####
2 #common setting
3 #国内外通用设置项 Common settings applicable both domestically and internationally
4 #####
5 LANGUAGE: 'en' 1
6 DEBUG_MODE: False #用户选择: zh:中国用户 en:International users #调试模式
7 #####
8 # dify setting
9 # dify配置选项
10 #####
11 DIFY_BASE_URL: 'http://localhost/v1'
12 #####
13 #####
14 ##### 中国用户 #####
15 # main-camera
16 DIFY_API_KEY: 'app-7bp4ipthR09kuZJkwY000VVP' #dify多智能体大模型应用API_KEY
17 # usb-camera
18 DIFY_API_KEY: 'app-hgv2JMH3KtBppuc42w13vme' #dify多智能体大模型应用API_KEY
19 #####
20 ##### International users #####
21 # main-camera
22 DIFY_API_KEY: 'app-S1273phJF71yPut4w3dROIn' #dify Multi-Agent Large Model Application API_KEY
23 # usb-camera
24 DIFY_API_KEY: 'app-ub1RKH0Cy21ldBQ6zF5uVvY' #dify Multi-Agent Large Model Application API_KEY
25 #####
26 #####
27 #####
28 #####
29 #ASR function setting
30 #语音识别功能设置
31 #####
32 ALIYUN_API_KEY: 'API_KEY' #阿里云API_KEY International users do not need to fill in this information.
33 USE_ONLINE_ASR: True #是否使用在线 ASR
34 ASR_SUPPLIER: 'xunfei' 3 #ASR供应商 (仅当使用在线ASR时生效): 中国大陆地区: aliyun 国际地区: xunfei #ASR Provider (Only applicable when using online ASR): Mainland China: aliyun; International
35 ONLINE_ASR_MODEL: 'parformer-realtime-v2'
36 ASR_THRESHOLD: 3 #ASR识别结果置信度, 低于该阈值则不进行后续处理, 单位: 字符 ASR recognition result threshold; if the result is below this threshold, no further processing is performed. 0
37 WAKEUP_THRESHOLD: 5.0 #唤醒时间阈值, 防止WAKEUP_THRESHOLD时间内的多次唤醒, 单位: 秒 wake-up time threshold to prevent multiple wake-ups within the WAKEUP_THRESHOLD time period, unit: sec
38 VAD_SENSITIVITY: 2 #语音灵敏度, 数值越高灵敏度越高, 可以关闭语音唤醒功能或调整灵敏度: VAD sensitivity
39 MAX_SILENCE_FRAMES: 90 #静音时长检测, 单位: 帧数, 1帧=10毫秒 End-of-sound duration detection, unit: frames, 1 frame = 10 milliseconds
40 #####
41 #####
42 #TTS function setting
43 #语音合成功能设置
44 #####
45 USE_ONLINE_TTS: True 4 #是否使用在线 TTS #Use online TTS?
46 TTS_SUPPLIER: 'xunfei' #TTS供应商 (仅当使用在线TTS时生效): 中国大陆地区: aliyun baidu 国际地区: xunfei #TTS providers (only applicable when using online TTS): Mainla
47 Baidu_API_KEY: 'API_KEY' #百度平台语音合成API_KEY
48 Baidu_SECRET_KEY: 'SECRET_KEY' #百度平台语音合成SECRET_KEY

```

- Other parameter modifications

`LANGUAGE` parameter: Domestic users should select zh, international users select en.

`DEBUG_MODE` parameter: Debug mode, when enabled you can see logs printed for each step in the terminal.

`USE_ONLINE_ASR` parameter: Select speech recognition method, can choose online or offline recognition. TRUE for online recognition (only Orin mainboard has offline recognition).

`ASR_SUPPLIER` parameter: Effective when online, domestic users need aliyun model, international users select xunfei model.

`USE_ONLINE_TTS` parameter: Select speech synthesis method, can choose online or offline recognition. TRUE for online recognition (only Orin mainboard has offline recognition).

`TTS_SUPPLIER` parameter: Effective when online, domestic users need aliyun, baidu models, international users select xunfei model.

6. Modify the Dfiy service API

- **Note:** This section is for reference only for users with development needs; it can generally be ignored.
- If you need to modify the API accessed by the car's infotainment system in the Dify application later, or if Dify is deployed on other servers, you will need to modify the address accessed by the car's infotainment system. Open the parameter configuration file:

```

gedit
~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/config/multi_brains_setting.
yaml

```

- Locate the `DIFY_API_KEY` and `DIFY_BASE_URL` parameters, where:

- `DIFY_BASE_URL` is the address for accessing the Dify backend service
- `DIFY_API_KEY` is the API key for the AI application in Dify

```
#####  
#   dify setting  
#   dify配置选项  
#####  
DIFY_BASE_URL: "http://localhost/v1"  
DIFY_API_KEY: "app-mhawRyoHteauIho7wvxhlhwr"           #dify applies  
API_KEY
```

7. Compiling the Function Package

Then, switch to the `/yahboomcar_ros2_ws/yahboomcar_ws#` workspace in the terminal:
Recompile the `multi_brains` function package to apply the configuration **(For Jetson Nano and Raspberry Pi , you need to enter Docker first)**

```
cd ~/yahboomcar_ros2_ws/yahboomcar_ws/
```

Recompile the function package:

```
colcon build --packages-select multi_brains
```

```
sunrise@ubuntu:~/yahboomcar_ros2_ws/yahboomcar_ws$ colcon build --packages-select multi_brains  
Starting >>> multi_brains  
Finished <<< multi_brains [10.6s]  
Summary: 1 package finished [12.2s]
```